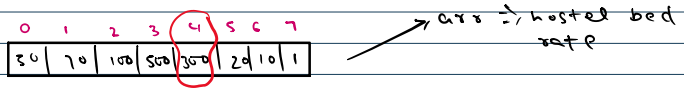
09 November 2025 13:32

Arrays

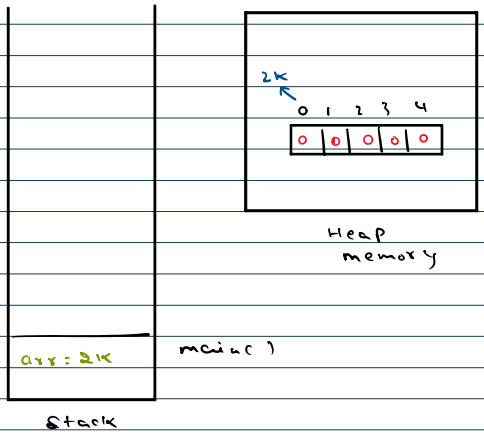
It is a non-primitive data structure of fixed size that stores multiple values of same data type in a single variable.

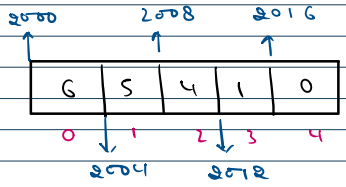
Each value in the array is accessed using an index, which starts from 0.



class → non-primitive → Heap memory

```java
public class Main {
    public static void main(String[] args) {
        int[] arr = new int[5];
    }

}
```





arr[2] → 4 (index)

↓

(base + 2*(size of data type) address)

↓

2K + 2*4

↓

2008

Base address + index * size of element

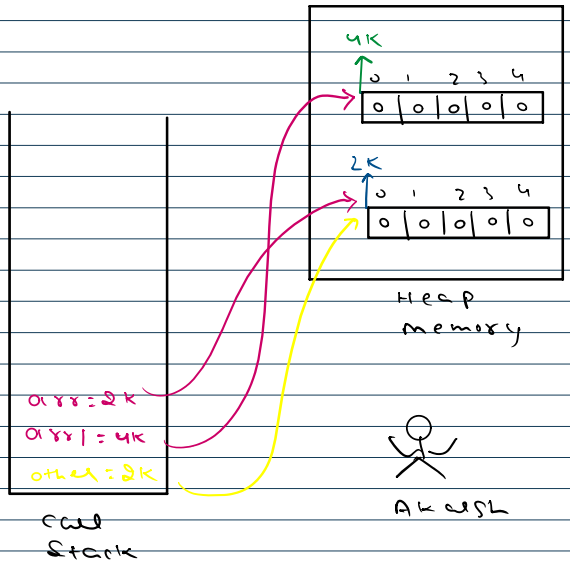
```java
public class Main {
    public static void main(String[] args) {
        int[] arr = new int[5];
        int arr1[] = new int[5]; // C Type
        System.out.println(arr);
        System.out.println(arr[0]);
        System.out.println(arr[1]);
        // System.out.println(arr[5]); // ArrayIndexOu

        int[] other = arr;
        System.out.println(other);
        System.out.println(other.length);

    }
}
```

Array operations

Get => int x = arr[2];

Set => arr[2] = 100;

```java
1  public class Main {
2      public static void main(String[] args) {
3          int[] arr = new int[5];
4          int arr1[] = new int[5]; // C Type
5          System.out.println(arr);
6          System.out.println(arr[0]);
7          System.out.println(arr[1]);
8          // System.out.println(arr[5]); // ArrayIndexOutOfBoundsException
9
10         int[] other = arr;
11         System.out.println(other);
12
13
14     }
15
16 }
```



tOfBoundsException

Taking User Input

0	1	2	3	4
10	20	30	40	50

i : 0 → arr[0] : 10 (user input)
i : 1 → arr[1] : 20
i : 2 → arr[2] : 30
i : 3 → arr[3] : 40
i : 4 → arr[4] : 50

```
for(int i=0; i<5; i++) {
    arr[i] = sc.nextInt();
}
```

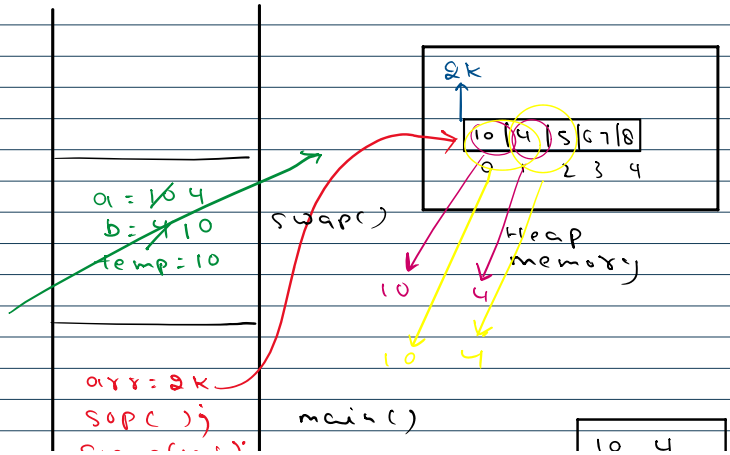
```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i=0; i<arr.length; i++) {
            arr[i] = sc.nextInt();
        }

        display(arr);
    }

    public static void display(int[] arr) {
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();
    }

}
```

Ways to declare

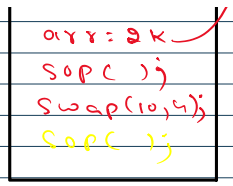
```
public class Main {
    public static void main(String[] args) {
        int[] arr1 = {10, 4, 5, 67, 8};
        int arr2[] = {10, 4, 5, 67, 8};
        int[] arr3 = new int[]{10, 4, 5, 67, 8};
        int arr4[] = new int[]{10, 4, 5, 67, 8};
        int[] arr5 = new int[5];
        int arr6[] = new int[5];
    }

}
```

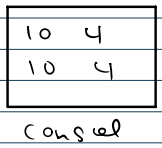
```
1  public class Main {
2      public static void main(String[] args) {
3          int[] arr = {10, 4, 5, 67, 8};
4          System.out.println(arr[0] + " " + arr[1]);
5          swap(arr[0], arr[1]);
6          System.out.println(arr[0] + " " + arr[1]);
7      }
8
9      public static void swap(int a, int b){
10         int temp = a;
11         a = b;
12         b = temp;
13     }
14
15 }
```

JVM



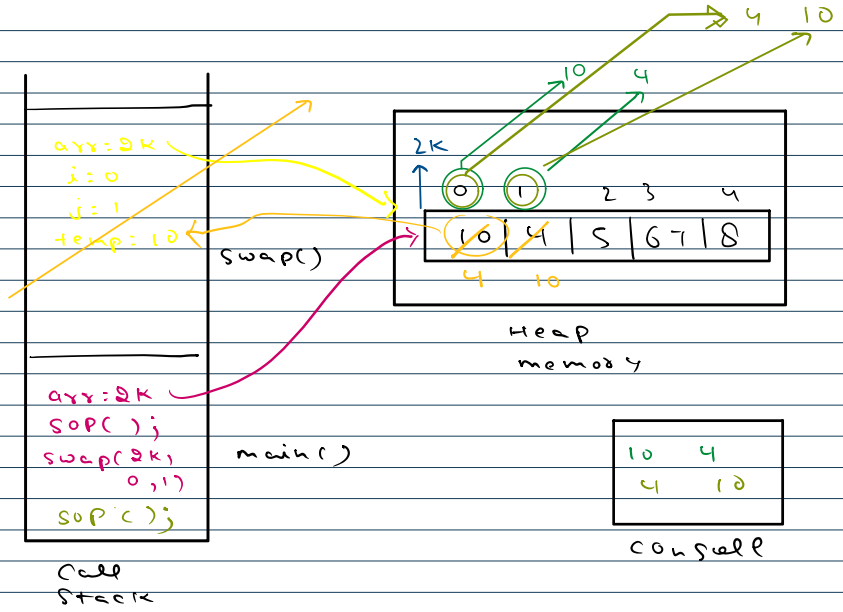
```
    }

}
```





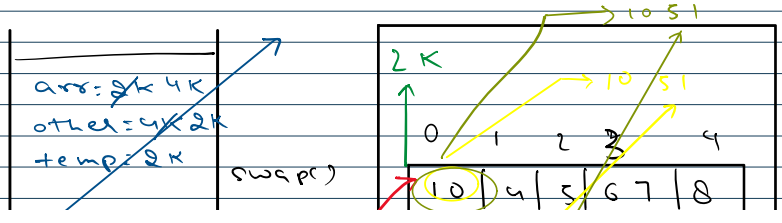

```
public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 4, 5, 67, 8};
        System.out.println(arr[0] + " " + arr[1]);
        swap(arr[0], arr[1]);
        System.out.println(arr[0] + " " + arr[1]);
    }

    public static void swap(int a, int b){
        int temp = a;
        a = b;
        b = temp;
    }

}
```

```
public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 4, 5, 67, 8};
        System.out.println(arr[0] + " " + arr[1]);
        swap(arr, 0, 1);
        System.out.println(arr[0] + " " + arr[1]);
    }

    public static void swap(int[] arr, int i, int j){
        int temp = arr[i];
        arr[i] = arr[j];
        arr[j] = temp;
    }
}
```

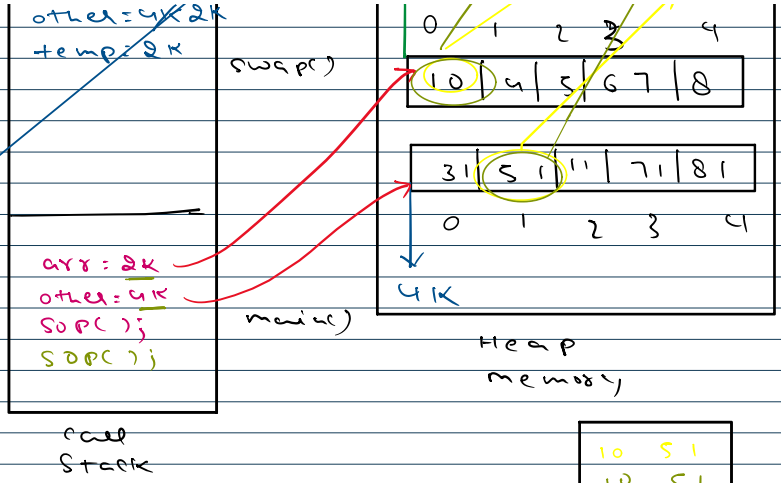



```
public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 4, 5, 67, 8};
        System.out.println(arr[0] + " " + arr[1]);
        swap(arr, 0, 1);
        System.out.println(arr[0] + " " + arr[1]);
    }

    public static void swap(int[] arr, int i, int j){
        int temp = arr[i];
        arr[i] = arr[j];
        arr[j] = temp;
    }

}
```

```
public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 4, 5, 67, 8};
        int[] other = {31, 51, 11, 71, 81};
        System.out.println(arr[0] + " " + other[1]);
        swap(arr, other);
        System.out.println(arr[0] + " " + other[1]);
    }
```



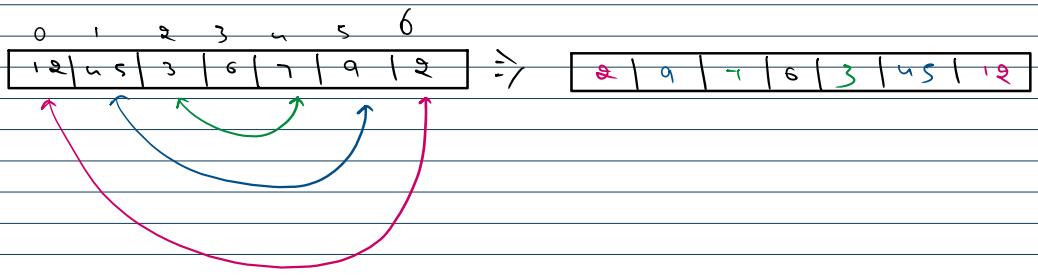
```java
        System.out.println(arr[0] + " " + other[1]);
        swap(arr, other);
        System.out.println(arr[0] + " " + other[1]);
    }

    public static void swap(int[] arr, int[] other){
        int[] temp = arr;
        arr = other;
        other = temp;
    }
}
```



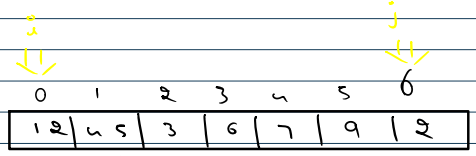
Reverse an array

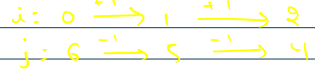
arr → 12, 4, 5, 3, 6, 7, 9, 2

reverse of arr → 2, 9, 7, 6, 3, 5, 4, 12



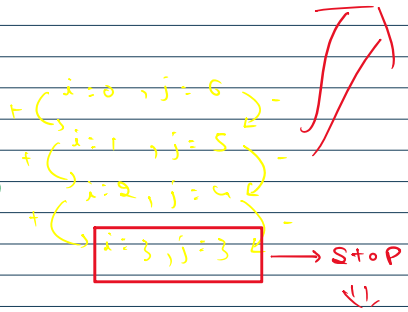
swap(arr[0], arr[6])
swap(arr[1], arr[5])
swap(arr[2], arr[4])





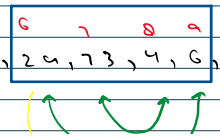
i < j ← pos

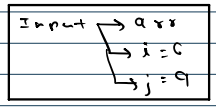
swap(arr[0], arr[6])
swap(arr[1], arr[5])
swap(arr[2], arr[4])



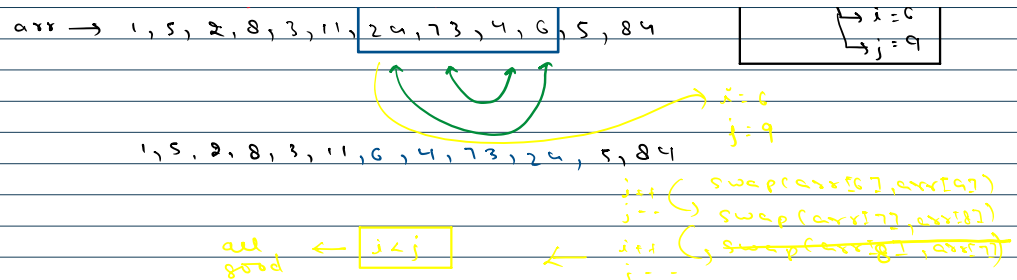
Reverse part of an array

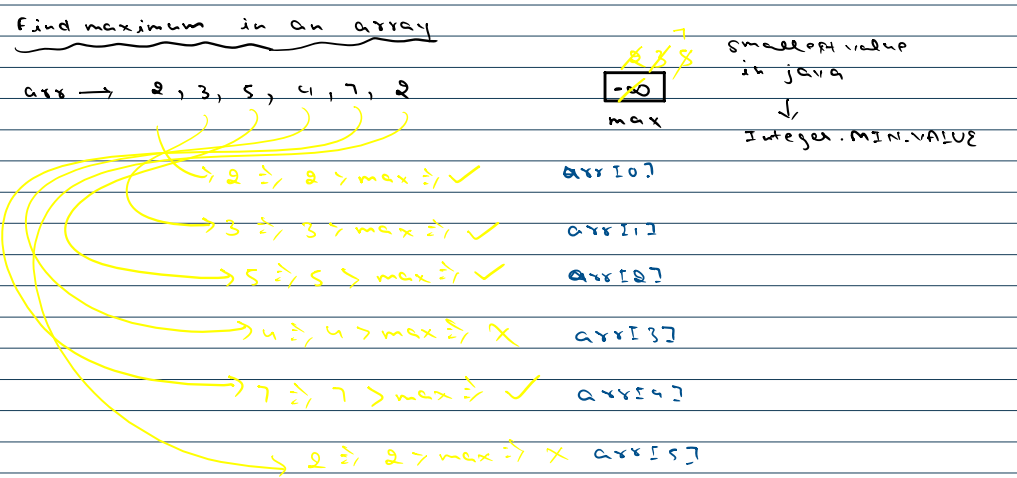
arr → 1, 5, 2, 8, 3, 11, 24, 73, 4, 6, 5, 84





i: 6





Find maximum in an array


```
public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 4, 5, 67, 8};
        System.out.println(maxValue1(arr));
        System.out.println(maxValue2(arr));

        System.out.println(Math.max(4, 5));
        // System.out.println(Math.max(4, 5, 89));
        System.out.println(Math.max(4, Math.max(5, 89)));
    }

    public static int maxValue1(int[] arr) {
        int max = Integer.MIN_VALUE;
        for(int i=0; i<arr.length; i++){
            if(arr[i] > max) {
                max = arr[i];
            }
        }
        return max;
    }

    public static int maxValue2(int[] arr) {
        int max = Integer.MIN_VALUE;
        for(int i=0; i<arr.length; i++){
            max = Math.max(arr[i], max);
        }
        return max;
    }

}
```